

Sandipan Borthakur

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h-index: 4 | Citations: 34 | First-authored papers: 3 | Co-authored papers: 5 |
Student advised: 4

EDUCATION

Tartu Observatory, University of Tartu (UT), Estonia; Graz University of Technology (TU Graz), Austria; Space Research Institute (IWF), Graz, Austria Doctor of Philosophy in Astronomy Thesis topic: Studying the impact of accretion on the composition of Herbig Ae/Be stars. Primary advisors: Mihkel Kama (UCL) & Luca Fossati (IWF)	<i>Feb. 2023 – Nov. 2026</i>
A&A Division, Physical Research Laboratory, India Junior Research Fellow Thesis: Photometric and spectroscopic studies of cataclysmic variables and classical novae.	<i>July 2019 – July 2021</i>
University of Delhi, India MSc. Physics	<i>July 2017 – June 2019</i> Overall Percentage: 66.5
Dayal Singh College, University of Delhi, India BSc Physics	<i>July 2013 – June 2016</i> Overall Percentage: 80.14

PUBLICATIONS

- [In preparation] “*Evolution of Oxygen abundance in Herbig Ae/Be stars*”, Borthakur, S. P. D., et al., Nature Astronomy, 2026
- “*The refractory fraction of phosphorus in planet-forming discs*”, Borthakur, S. P. D., et al., A&A Letters, 2026
- “*The Tartu Observatory Fiber-fed Echelle Spectrograph (TOFES) Data Reduction Pipeline*”, Borthakur, S. P. D., et al., PASP, 2026
- “*Abundance analysis of stars hosting gas-rich debris discs*”, Borthakur, S. P. D., et al., A&A, 2025
- “*Refractory phosphorus in the HD 100546 protoplanetary disc*”, Kama, M., Shorttle, O., Borthakur, S. P. D., et al., MNRAS, 2025
- “*Ariel stellar characterisation [IV] : Fundamental Parameters of 18 Hot Stars in the Ariel Mission Candidate Sample*”, Ramler, H., Borthakur, S. P. D., et al., A&A, 2026
- “*Non-local thermodynamical equilibrium atmospheric modelling of the ultra-hot Jupiter WASP-178b and comparison with UV and optical observations*”, Fossati, L., et al. (incl. Borthakur, S. P. D.), A&A, 2025
- “*Atmospheric dynamics of the hypergiant RW Cep during the Great Dimming*”, Kasikov, A., et al. (incl. Borthakur, S. P. D.), A&A, 2025

NON-REFEREED PUBLICATIONS

Spectroscopic classification of ASASSN-21fh as a classical nova of the Fe II class, Joshi V., et al. (incl. **Borthakur, S. P. D.**), The Astronomer’s Telegram, 14544, 1, 2021

Dust formation in V1112 Per (Nova Per 2020); Banerjee, D. P. K., et al. (incl. **Borthakur, S. P. D.**), The Astronomer’s Telegram, 14338, 1, 2021

“Reverse hybrid” behavior of Nova Per 2020 = TCP J04291888+4354233, **Borthakur S.**, et al., The Astronomer’s Telegram, 14230,1, 2020

CONFERENCES AND SEMINARS

University of Wisconsin-Madison Monday Seminar talk: “Tracing the inner protoplanetary disc composition using Herbig star photospheres”, 26 October, 2026.

Contributed talk: “Tracing inner protoplanetary disk composition from stellar photospheres of Herbig Ae/Be stars”, **European Astronomical Society meeting**, Switzerland, 29 June–03 July, 2026.

Contributed talk: “What do main-sequence A-type stars tell us about their earlier planet-forming disks?”, **European Astronomical Society meeting**, Switzerland, 29 June–03 July, 2026.

Contributed talk: “Linking Herbig Ae/Be stellar composition to inner disk composition: Constraining the elemental budget in the inner disk”, **European Astronomical Society meeting**, Switzerland, 29 June–03 July, 2026.

Contributed talk: “Stellar accretion contamination: Protoplanetary to Debris disks”, **Exploring star-planet-disk connections** conference, Estonia, 14–17 October 2025.

Contributed talk: “Phosphorus and Oxygen inner disk composition from stellar contamination”, **Graz-Vienna Exoplanet Scientist Meeting VI**, Austria, 23 May, 2025

Invited talk: “Accretion contamination on the photosphere of Herbig stars”, **COST Action PLANETS WG3 Science seminar**, 18 November, 2025.

Poster presentation: “Photospheric contamination in early-type stars: From protoplanetary to debris disks”, **European Astronomical Society Meeting**, Ireland, 23–27 June, 2025.

Poster presentation: “Stellar Photospheric Contamination: Evolution from Protoplanetary to Debris Disks”, **ESO’s New Height in Planet formation** conference, Germany, 15–19 July, 2024.

AWARDS AND GRANTS

Young Researcher and Innovator Conference Grant from COST Action PLANETS for the EAS 2026, Lausanne, Switzerland conference visit. (EUR 1400)

Ernst Julius Öpik scholarship, 2025 from the University of Tartu. (EUR 1200)

CSIR-UGC National Eligibility Test (NET), 2019, Physical Sciences – Qualified for Assistant Professorship in India.

OBSERVING PROPOSAL

(Accepted) Co-I: APEX proposal “*Testing the universality of proto-planetary phosphorus reservoirs*”.

(Submitted) PI: ESO P118 observing proposal 2026, “*Impact of earlier protoplanetary disc accretion on young main sequence A-type stars*”.

(Submitted) co-PI: JWST MIRI Cycle 6 proposal 2026, “*Investigating the Link Between Herbig Disk Evolution and Photospheric Contamination*”.

SKILLS

Photometric and spectroscopic observing experience with Tartu observatory 1.5m and Mt. Abu 1.2m telescopes.

Zeeman spectral synthesis code for stellar parameter estimates.

PyReduce data reduction package for Echelle spectrographs.

chemcomp 1D protoplanetary disk evolution code.

Module for Experiment in Stellar Astrophysics (MESA) stellar evolution code.

Python, emcee, PyMultiNest, C++ programming, and software development using PyQt.

Familiar with a code which simulates exospheres of exoplanets (I am checking for the feasibility of exosphere detection using the Pollux instrument).

SUPERVISION, OUTREACH AND OTHER ACTIVITIES

Bachelor thesis supervisor 2025-2026: Supervised a Bachelor student from the Computer Science department, University of Tartu, for two semesters to develop a Graphical User Interface for the ZEEMAN spectral synthesis code.

High school students visit Tartu Observatory 2023-2025: Supervised a high-school student for almost two years on stellar spectroscopic analysis. She is a co-author of Ramler et al. 2026.

High school students visit Tartu Observatory 2024: Supervised a high school student for a month on studying the phosphorus abundance in Herbig Ae/Be stars.

Seminar on the Scientific Method 2023: Supervised a physics Bachelor’s student for a semester in studying and presenting a published paper in a seminar to develop their ability to understand scientific literature.

Starry Nights in Tõravere volunteer 2023–2025 (multiple nights over the year): Open-doors event. I volunteered to operate the Tartu Observatory telescope for visitors.

Science Express, Physical Research Laboratory, India 2019–2021: Outreach activity in different schools and colleges to showcase fun and interactive science experiments for the public.

Member of Local Organising Committees: “Asteroid Research Training Workshop”, Tõravere, Estonia, 21–25 August 2023.

Member of Local Organising Committees: “Exploring star-planet-disk connections” conference, Estonia, 14–17 October 2025.

Member of Local Organising Committees: “Ariel Consortium Meeting”, Estonia, 23–26 April 2024.

Member of Local Organising Committees: “Exohost Stellar Spectroscopy Workshop”, Estonia, 27–28 September 2023.

Member of LOC and Scientific Organising Committee: “Exohost Planet formation and planet populations” summer school, Estonia, 16–20 September 2024.

MEMBERSHIP

Member of the Stellar Characterisation Working Group of the Ariel space mission consortium.

Member of the Pollux science consortium. Pollux is one of the proposed science instruments for the Habitable Worlds Observatory.

Member of the Waltzer consortium. Waltzer is a proposed F-class ESA mission to observe and characterise exoplanet atmospheres.